

Task 23 — Hardening, Scale & MLOps

Data Analyst Submission Report

Objective

Build a decision-grade analytics setup in which every metric can be traced to a source event, validated for data quality and freshness, and used to support an operational decision. The focus is hardening, scale, failure handling, and MLOps readiness.

Dataset

The accompanying CSV contains 36 validated event/metric records across application, document, analytics-pipeline, and dashboard services. It includes metric source, data-quality status, freshness checks, dependency checks, edge-case handling, and decision-support fields.

Metric dictionary

Metric	Meaning	Decision use
Events processed	Number of events handled by the pipeline	Detect scale/load changes
Success rate %	Successful events as a percentage of processed events	Monitor reliability
Data freshness min	Minutes between event availability and dashboard/report data	Ensure timely data
Error rate %	Share of events producing an error	Trigger investigation/recovery

Validation and findings

- All 36 records passed the data-quality check.
- All records passed the freshness check and dependency check.
- Two edge-case records were explicitly marked as HANDLED, showing that failures are treated as expected operational conditions rather than hidden problems.
- The metrics are source-labelled, making each dashboard number traceable to validated event data.
- The structure supports an MLOps-style operating model: monitor → validate → detect failure → recover → document evidence → make a decision.

Decision-grade conclusion

The pipeline is suitable for a hardened analytics dashboard because its metrics have explicit definitions, source lineage, freshness and dependency checks, and visible edge-case handling. The next operational step is to connect these validated metrics to live alerts and a dashboard so that a founder or operator can act on changes without relying on unexplained numbers.

Submission evidence

CSV: Task_23_Hardening_Scale_MLOps.csv

Recommended dashboard cards: Events Processed, Success Rate, Data Freshness, Error Rate, Dependency Health, and Edge-Case/Failure Status.